

CUSTOMER CHURN PREDICTION & LIFETIME VALUE (LTV) ENGINE

Production-Level Data Analytics & Predictive Analytics Project

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Course: B.Tech – Computer Science & Engineering

Project Type: Data Analytics & Machine Learning

Domain: Telecommunications / Subscription Analytics

1. PROJECT OVERVIEW

The Customer Churn Prediction & Lifetime Value (LTV) Engine is an end-to-end data analytics and predictive analytics project designed for telecommunications and subscription-based businesses.

The project analyzes customer data to understand customer behavior, identify churn patterns, predict customer churn risk, analyze Customer Lifetime Value (LTV), and segment customers based on their value.

The project combines Python, SQL, PostgreSQL, SQLAlchemy, Machine Learning, FastAPI, Docker, and Metabase to create a complete analytics and prediction workflow.

The system provides both single-customer and batch customer churn prediction through REST API endpoints.

2. BUSINESS PROBLEM

Customer churn is a major challenge for telecommunications and subscription-based businesses.

When customers leave a service, businesses lose recurring revenue and may need to spend additional resources to acquire new customers.

Therefore, identifying customers who are likely to churn can help businesses take proactive retention actions.

This project addresses the problem by analyzing customer information such as:

- Customer tenure
- Monthly charges
- Total charges
- Contract information
- Internet service
- Payment method
- Customer service usage
- Churn status
- Customer Lifetime Value

The project also provides LTV segmentation to understand customer value.

3. PROJECT OBJECTIVES

The main objectives of the project are:

- Analyze customer churn patterns.
- Understand factors associated with customer churn.
- Predict whether a customer is likely to churn.
- Generate churn probability.
- Analyze Customer Lifetime Value.
- Segment customers based on LTV.
- Identify customers with higher churn risk.
- Perform retention-related analysis.
- Store and manage analytical data using PostgreSQL.
- Develop a REST API using FastAPI.
- Support single-customer prediction.
- Support batch customer prediction.
- Containerize the API using Docker.
- Build interactive Metabase dashboards.
- Provide data-driven insights for customer retention.

4. DATA SOURCE

The primary dataset used in this project is the Telco Customer Churn Dataset.

The dataset contains approximately 7,043 customer records and includes customer demographic, service, contract, billing, and churn-related information.

Dataset File

- TELCO_CHURN.CSV.xlsx
- Important Dataset Fields
- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies
- Contract
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges

- Churn
- LTV

Figure 1: Telco Customer Churn Dataset

Customer Churn DB / Customer Ltv Features

Customer ID	Tenure	Monthly Charges	Total Charges	Contract	Internet Service	Payment Method	Churn	
7590-VHVEG	1	29.85	29.85	Month-to-month	DSL	Electronic check	No	29.85
5575-GNVDE	34	56.95	1,889.5	One year	DSL	Mailed check	No	1,936.3
3668-QPYBK	2	53.85	108.15	Month-to-month	DSL	Mailed check	Yes	107.7
7795-CFOCW	45	42.3	1,840.75	One year	DSL	Bank transfer (automatic)	No	1,903.5
9237-HQITU	2	70.7	151.65	Month-to-month	Fiber optic	Electronic check	Yes	141.4
9305-CDSKC	8	99.65	820.5	Month-to-month	Fiber optic	Electronic check	Yes	797.2
1452-KIOVK	22	89.1	1,949.4	Month-to-month	Fiber optic	Credit card (automatic)	No	1,960.2
6713-OKOMC	10	29.75	301.9	Month-to-month	DSL	Mailed check	No	297.5
7892-POOKP	28	104.8	3,046.05	Month-to-month	Fiber optic	Electronic check	Yes	2,934.4
6388-TABGU	62	56.15	3,487.95	One year	DSL	Bank transfer (automatic)	No	3,481.3
9763-GRSKD	13	49.95	587.45	Month-to-month	DSL	Mailed check	No	649.35
7469-LKBCI	16	18.95	326.8	Two year	No	Credit card (automatic)	No	303.2

Showing first 2,000 rows 296ms

Customer Churn DB / Customers

Customer ID	Gender	Senior Citizen	Partner	Dependents	Tenure	Phone Service	Multiple Lines	Internet Service	
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No
6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No
6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes
9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes
7469-LKBCI	Male	0	No	No	16	Yes	No	No	No inte

Showing first 2,000 rows 282ms

Customer ID	Gender	Senior Citizen	Partner	Dependents	Tenure	Phone Service	Multiple Lines	Internet Service	
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No
9305-COSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No
6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No
6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes
9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes
7469-LKBCI	Male	0	No	No	16	Yes	No	No	No inte

5. TECHNOLOGY STACK

- Programming Languages
- Python
- SQL
- Data Analysis
- Pandas
- NumPy
- Machine Learning
- Scikit-learn
- XGBoost
- SHAP
- Database
- PostgreSQL
- SQLAlchemy
- API Development
- FastAPI
- Pydantic
- Uvicorn
- Swagger UI
- Deployment
- Docker
- Business Intelligence
- Metabase
- Development Tools
- Visual Studio Code
- Python Scripts / Notebooks

6. PROJECT ARCHITECTURE

The overall project workflow is:

Telco Customer Churn Dataset



Data Cleaning & Preprocessing



Exploratory Data Analysis



Churn Analysis



Feature Engineering



Churn Prediction & LTV Analysis



PostgreSQL / SQLAlchemy



FastAPI



Swagger API Testing



Docker Deployment

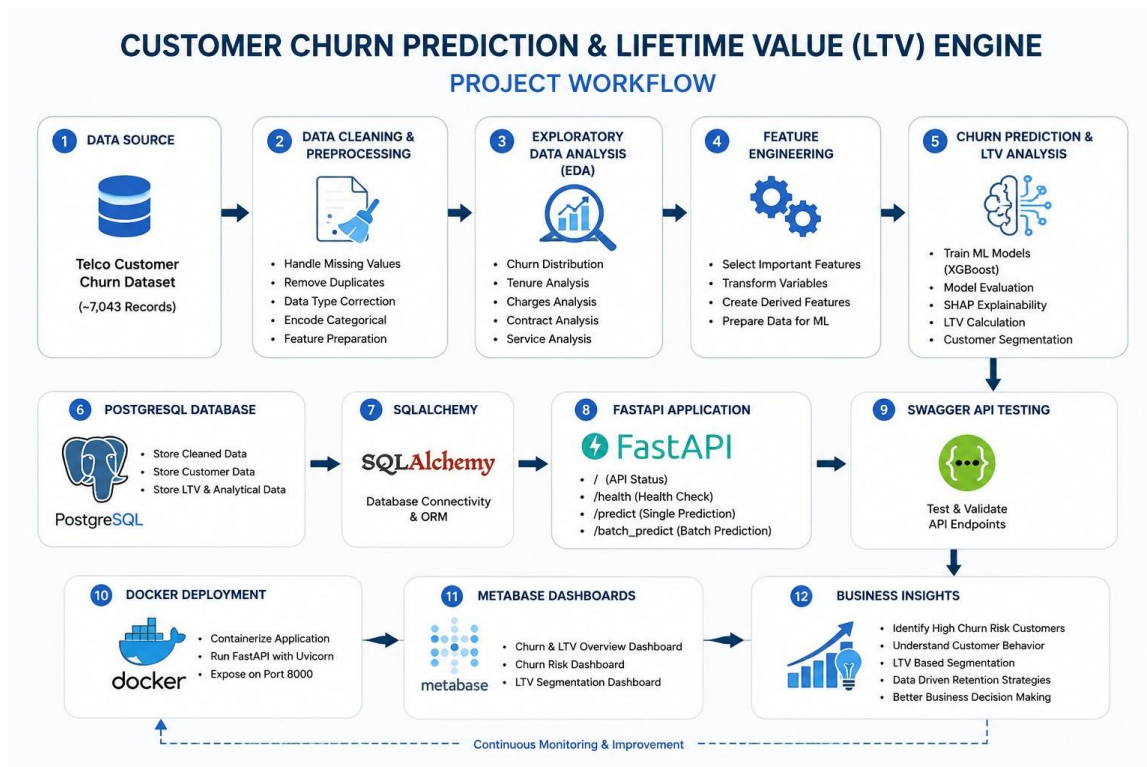


Metabase



Business Insights

Figure 2: Project Architecture / Workflow



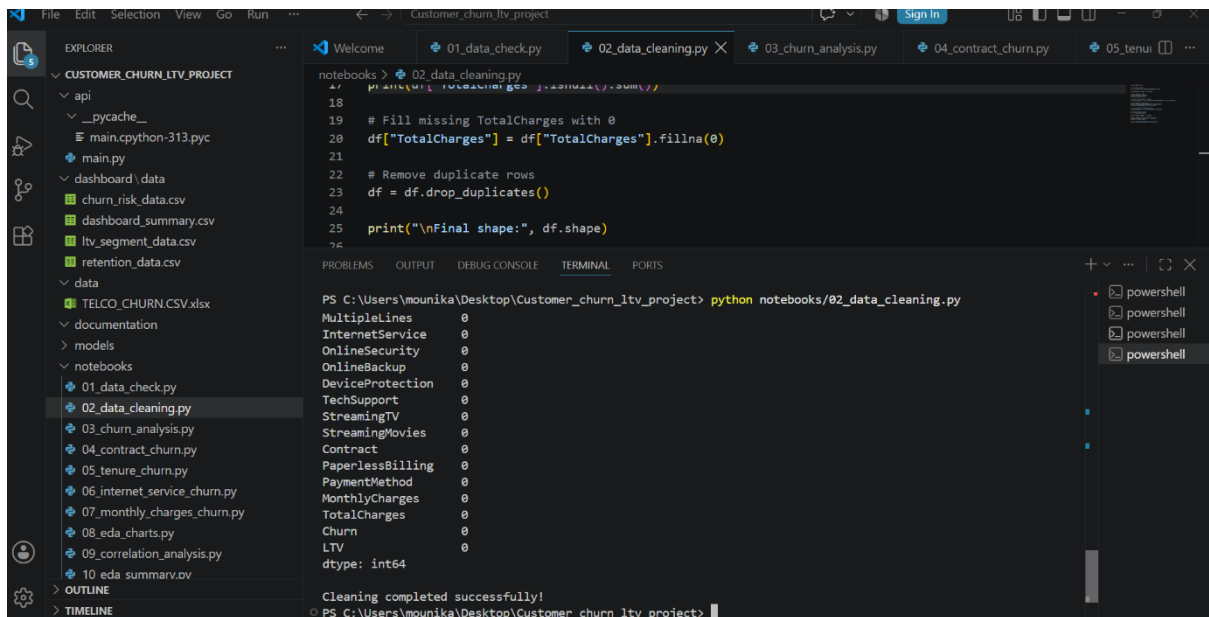
7. DATA CLEANING AND PREPROCESSING

The raw Telco Customer Churn dataset was inspected and prepared before analysis.

The data preparation process included:

- Inspecting the dataset structure.
- Checking missing values.
- Checking duplicate records.
- Validating data types.
- Cleaning numerical fields.
- Preparing categorical variables.
- Preparing customer-related analytical fields.
- Preparing data for churn analysis.
- Preparing data for LTV analysis.
- The cleaned data was then used for exploratory analysis and predictive analytics.

Figure 3: Data Cleaning and Preprocessing



8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand customer behavior and identify churn-related patterns.

The analysis focused on:

- Customer tenure
- Monthly charges
- Total charges
- Contract type
- Internet service
- Customer service usage
- Payment method
- Churn distribution
- Customer value
- Major EDA Analysis
- Churn distribution analysis
- Churn by contract type
- Churn by tenure
- Churn by monthly charges
- Churn by internet service
- Customer behavior analysis
- LTV-related analysis

EDA helped identify important customer patterns that could be used for churn-risk and retention analysis.

9. FEATURE ENGINEERING

Feature engineering was performed to prepare meaningful customer attributes for churn and LTV analysis.

Important prediction variables include:

- Tenure
- Monthly Charges
- Total Charges

Additional analytical data was prepared for:

- Churn risk analysis
- LTV analysis
- Customer segmentation
- Retention analysis

Feature engineering helped transform customer information into useful inputs for predictive and analytical tasks.

10. CHURN PREDICTION

The project includes a customer churn prediction component.

The prediction API uses customer information including:

- Tenure
- Monthly Charges
- Total Charges

The API generates:

- Churn Prediction
- Churn Probability

The prediction indicates whether the customer is predicted to churn, while the probability represents the estimated churn risk.

11. CUSTOMER LIFETIME VALUE (LTV) ANALYSIS

Customer Lifetime Value analysis is used to understand the long-term value of customers.

The project analyzes customer value and creates LTV-based segments.

LTV Analysis Includes :

- Customer LTV
- LTV distribution
- Customer value segmentation
- High-value customers
- Medium-value customers
- Low-value customers

LTV segmentation helps businesses understand customer value and prioritize customer retention efforts.

12. CHURN RISK AND RETENTION ANALYSIS

The project includes churn-risk and retention-related analysis.

The analysis focuses on identifying customers who may have a higher risk of leaving the service.

Important areas include:

- Churn probability
- Churn risk
- Customer tenure
- Monthly charges
- Customer value
- Retention-related customer groups

The combination of churn risk and customer value can support targeted retention strategies.

13. POSTGRESQL DATABASE

PostgreSQL was used as the database technology in the project.

The database environment was used to manage customer and LTV-related analytical data.

Main Tables

- customers
- Itv

SQLAlchemy was used as the database connectivity layer.

PostgreSQL supports the structured data management and analytical workflow of the project.

> Note: The current FastAPI /predict implementation returns predictions directly; it does not automatically save each API prediction into PostgreSQL.

14. FASTAPI APPLICATION

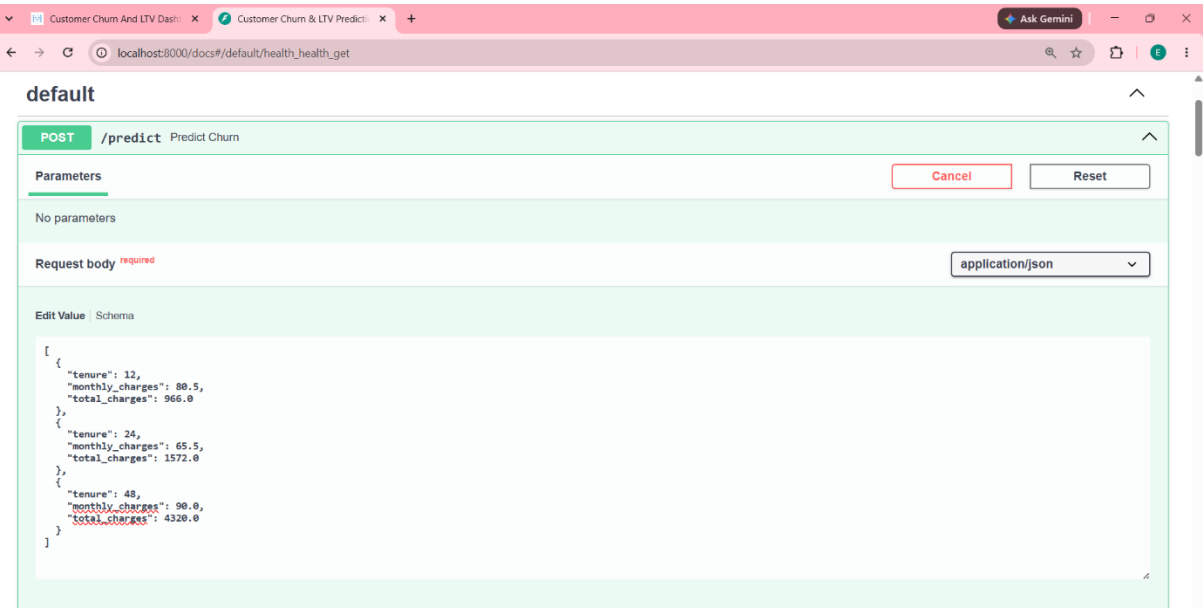
FastAPI was used to develop the REST API for customer churn prediction.

API Endpoints

Endpoint	Method	purpose
/	GET	Check API status
/health	GET	Check API health
/predict	POST	Predict churn for one customer
/batch_predict	POST	Predict churn for multiple customer

Swagger UI was used to test and verify the API endpoints.

Figure 5: FastAPI Swagger UI



15. SINGLE CUSTOMER API TESTING

The /predict endpoint was successfully tested using sample customer information.

Input

parameter	value
Tenure	12
Monthly charges	80.5
Total Charges	966.0

Output

Churn Prediction: Yes

Churn Probability: 0.60 / 60%

HTTP Status: 200 OK

This confirms that the API successfully processed the customer information and returned a prediction.

Figure 6: /predict API Successful Response

Customer Churn & LTV Prediction API

1.0.0OAS 3.1

/openapi.json

default

POST /predict Predict Churn

Parameters

No parameters

Request body required

application/json

Edit Value Schema

```
{  "tenure": 12,  "monthly_charges": 80.5,  "total_charges": 966.0}
```

Responses

Curl

```
curl -X 'POST' \  'http://localhost:8000/predict' \  -H 'accept: application/json' \  -H 'Content-Type: application/json' \  -d '{  "tenure": 12,  "monthly_charges": 80.5,  "total_charges": 966.0  }'
```

Request URL

http://localhost:8000/predict

Server response

Code

Details

200

Response body

```
{  "churn_prediction": "Yes",  "churn_probability": 0.6}
```

Response headers

```
content-length: 50  content-type: application/json  date: Wed, 12 Aug 2026 17:57:02 GMT  server: uvicorn
```

Responses	
Code	Description
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>"string"</pre>
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>

16. BATCH PREDICTION API

The /batch_predict endpoint was tested using three customers simultaneously.

Input Data

Customer	Tenure	Monthly charges	Total charges
Customer 1	12	80.5	966.0
Customer 2	24	65.5	1572.0
Customer 3	48	90.0	4320.0

Results

Customer	Churn Prediction	Churn Probability
Customer1	yes	60%
60%Customer2	No	20%
Customer3	No	20%

Total Customers Processed: 3

HTTP Status: 200 OK

Figure 7: /batch_predict API Successful Response

POST /batch_predict Batch Predict

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
[
  {
    "tenure": 12,
    "monthly_charges": 80.5,
    "total_charges": 966.0
  },
  {
    "tenure": 24,
    "monthly_charges": 65.5,
    "total_charges": 1572.0
  },
  {
    "tenure": 48,
    "monthly_charges": 90.0,
    "total_charges": 4320.0
  }
]
```

Execute

Clear

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string" No links	
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre> No links	

17. API HEALTH CHECK

The /health endpoint was tested to verify that the API service was running correctly.

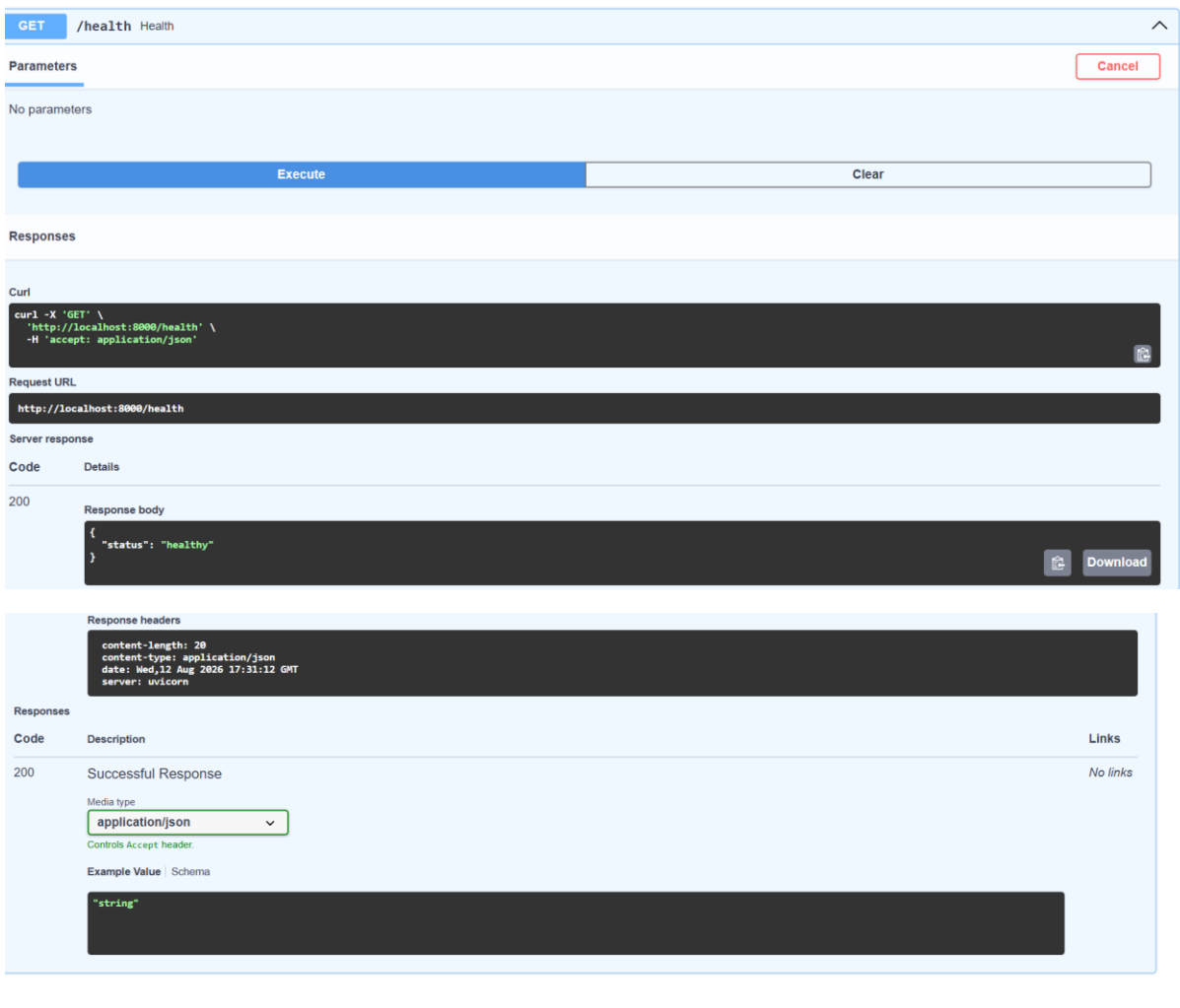
Response

```
{
  "status": "healthy"
}
```

HTTP Status: 200 OK

This confirms that the FastAPI service was healthy and operational.

Figure 8: API Health Check



18. DOCKER DEPLOYMENT

Docker was used to containerize and run the FastAPI application.

The Docker container successfully ran the API application and exposed it through port 8000.

Deployment Flow

FastAPI Application



Docker Image



Docker Container



Uvicorn Server



Port 8000

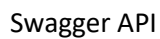


Figure 9: Docker Container Running



Three dashboards were developed:

1. Churn & LTV Overview
2. Churn Risk
3. LTV Segmentation

The dashboards use KPI cards, charts, and analytical visualizations to provide customer churn and LTV insights.

20. DASHBOARD 1 – CHURN & LTV OVERVIEW

The Churn & LTV Overview Dashboard provides a high-level summary of customer churn and Lifetime Value.

KPI Cards

The dashboard contains KPI cards related to:

Customer overview

Churn performance

Customer value

LTV-related metrics

Charts and Analysis

The dashboard provides visual insights into:

Customer churn

Customer value

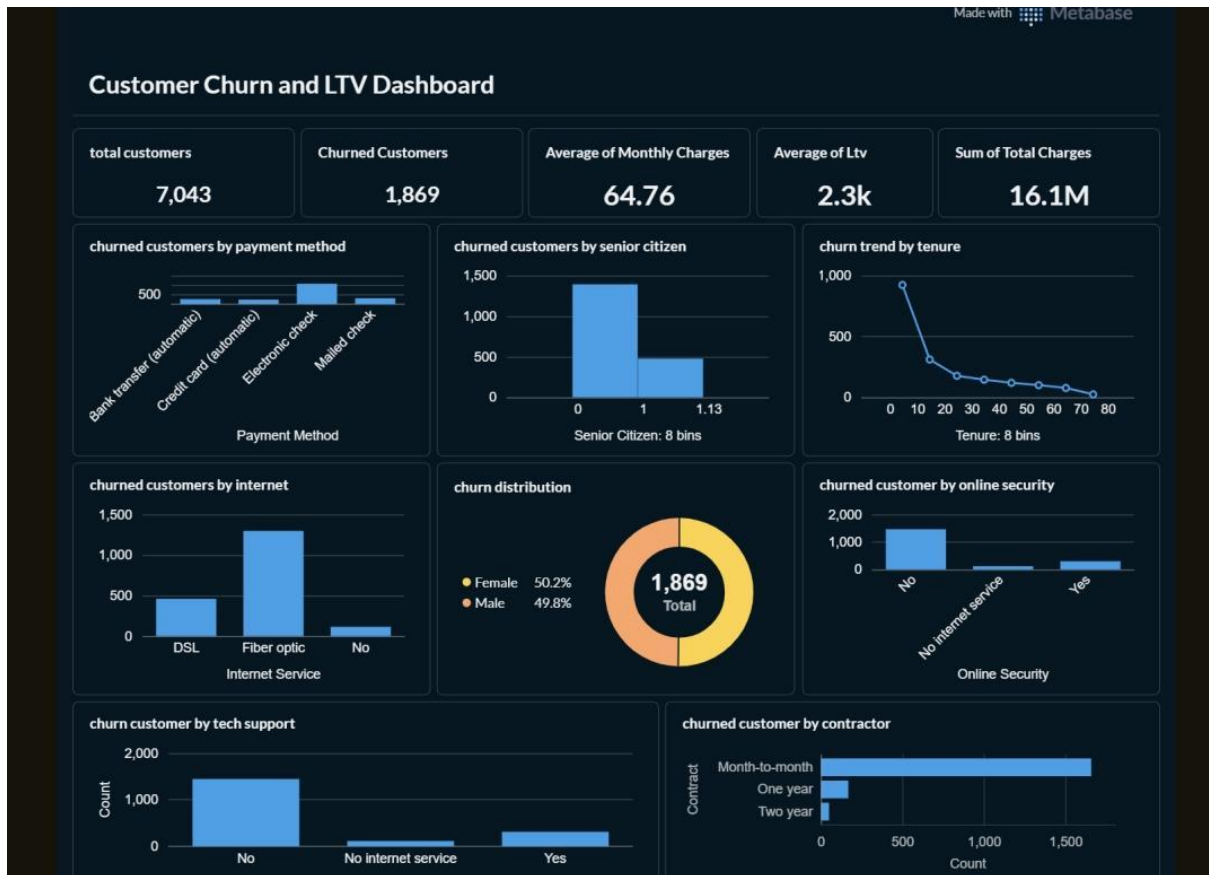
LTV distribution

Customer segments

Overall customer analysis

This dashboard provides an executive-level overview of customer churn and LTV.

Figure 10: Churn & LTV Overview Dashboard



21. DASHBOARD 2 – CHURN RISK

The Churn Risk Dashboard focuses on identifying customers who are at risk of churn.

KPI Cards

The dashboard provides churn-risk-related KPI cards.

Charts and Analysis

The dashboard analyzes:

- Churn risk
- Churn distribution
- Churn by tenure
- Churn by monthly charges
- Customer churn patterns
- High-risk customer groups

This dashboard helps businesses identify customers who may require retention attention.

Figure 11: Churn Risk Dashboard



22. DASHBOARD 3 – LTV SEGMENTATION

The LTV Segmentation Dashboard focuses on customer Lifetime Value.

KPI Cards

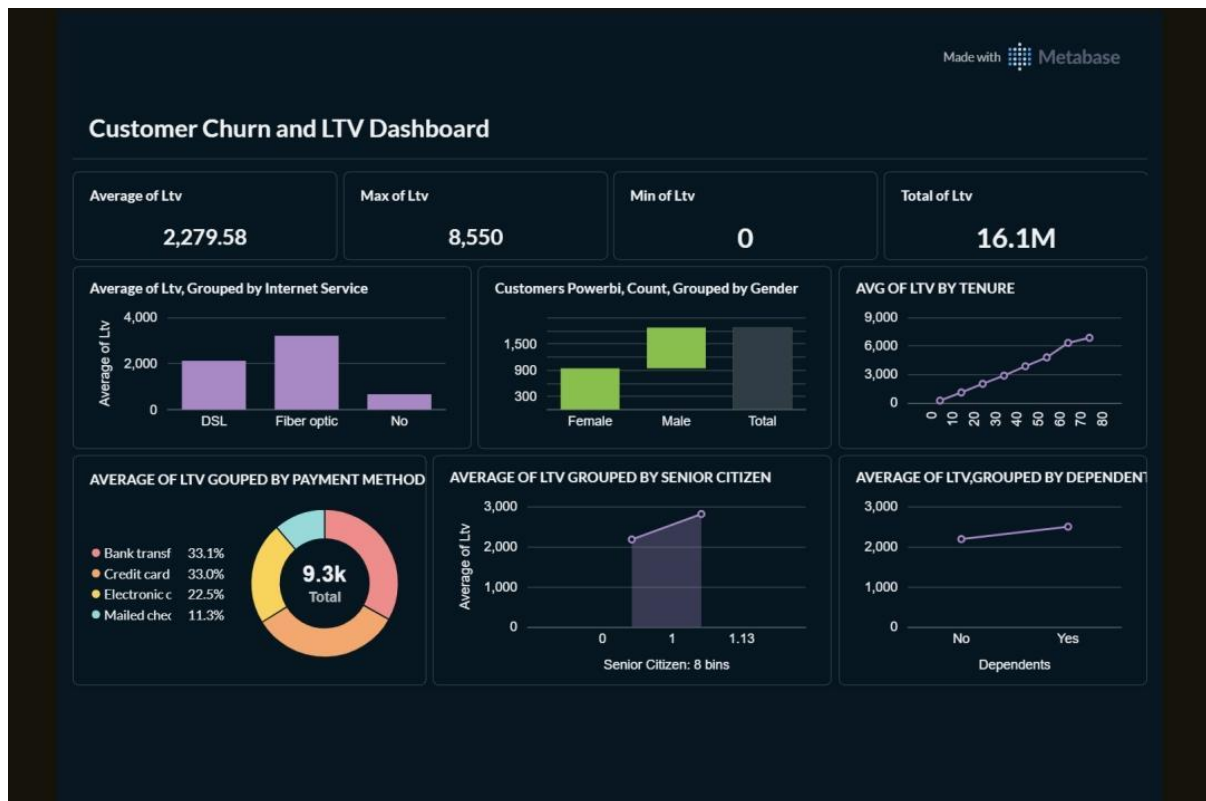
The dashboard provides LTV-related KPI cards.

Charts and Analysis

- The dashboard provides:
- LTV segmentation
- Customer value distribution
- High-value customer analysis
- Medium-value customer analysis
- Low-value customer analysis

This dashboard helps businesses understand customer value and prioritize valuable customer groups.

Figure 12: LTV Segmentation Dashboard



23. DASHBOARD DATA FILES

The project contains dedicated datasets used for dashboard analysis.

churn_risk_data.csv

Used for:

- Churn-risk analysis
- Customer risk analysis
- Churn-related visualizations
- dashboard_summary.csv

Used for:

- Dashboard summary information
- KPI-related analysis
- ltv_segment_data.csv

Used for:

- LTV segmentation
- Customer value analysis
- retention_data.csv

Used for:

- Retention analysis

- Customer retention-related insights

24. PROJECT STRUCTURE

CUSTOMER_CHURN_LTV_PROJECT/

- └─ api/
- | └─ __pycache__/
- | | └─ main.cpython-313.pyc
- | └─ main.py
- └─ dashboard/data/
- | └─ churn_risk_data.csv
- | └─ dashboard_summary.csv
- | └─ ltv_segment_data.csv
- | └─ retention_data.csv
- └─ data/
- | └─ TELCO_CHURN.CSV.xlsx
- └─ documentation/
- └─ models/
- └─ notebooks/
- | └─ 01_data_check.py
- | └─ 02_data_cleaning.py
- | └─ 03_churn_analysis.py
- | └─ 04_contract_churn.py
- | └─ 05_tenure_churn.py
- | └─ 06_internet_service_churn.py
- | └─ 07_monthly_charges_churn.py
- | └─ 08_eda_charts.py
- | └─ 09_correlation_analysis.py
- | └─ 10_eda_summary.py
- | └─ 11_feature_engineering.py

- | ├── 12_categorical_encoding.py
- | ├── 13_train_test_split.py
- | ├── 14_feature_scaling.py
- | ├── 15_logistic_regression.py
- | ├── 16_confusion_matrix.py
- | ├── 17_model_metrics.py
- | ├── 18_roc_auc.py
- | ├── 19_model_comparison.py
- | ├── 20_feature_importance.py
- | ├── 21_best_model_selection.py
- | ├── 22_final_churn_prediction.py
- | ├── 23_business_insights.py
- | ├── 24_save_final_model.py
- | ├── 25_xgboost_model.py
- | └── 26_shap_explainability.py
- |
- | ├── outputs/
- |
- | ├── src/
- | ├── 41_import_data_to_postgresql.py
- | ├── 41_sqlalchemy_pipeline.py
- | ├── 42_ltv_feature_engineering.py
- | ├── 42_verify_postgresql.py
- | ├── 43_ltv_regression_model.py
- | ├── 43_sql_business_analysis.py
- | ├── 44_churn_ltv_segmentation.py
- | ├── 44_store_prediction.py
- | ├── 45_retention_priority_scoring.py
- | ├── 50_dashboard_...py
- | └── database.py

└─ business_insights.csv
└─ Dockerfile
└─ feature_importance.png
└─ final_churn_predictions.csv
└─ requirements.txt
└─ .gitignore

25. FOUR-WEEK DEVELOPMENT TIMELINE

WEEK 1 – Data Ingestion & Exploratory Data Analysis

Day 1–2

- Set up PostgreSQL.
- Load the Telco Customer Churn dataset.
- Inspect the dataset structure.
- Check customer records and data quality.

Day 3–5

- Perform Exploratory Data Analysis.
- Analyze customer tenure.
- Analyze monthly charges.
- Analyze total charges.
- Analyze churn distribution.
- Analyze contract-related churn.
- Analyze internet-service-related churn.

Day 6–7

- Check missing values.
- Clean numerical fields.
- Validate categorical fields.
- Prepare the analytical dataset.

WEEK 2 – Feature Engineering & Predictive Analytics

Day 1–3

- Prepare churn-related features.
- Analyze tenure.
- Analyze monthly charges.
- Analyze total charges.
- Prepare prediction inputs.

Day 4–6

- Develop churn prediction functionality.
- Analyze churn probability.
- Perform feature analysis.

- Perform model-related analysis and evaluation.

Day 7

- Finalize churn-related analytical features.
- Prepare prediction results for API testing.

WEEK 3 – LTV Analysis & API Development

Day 1–3

- Perform LTV analysis.
- Prepare LTV segments.
- Prepare churn-risk data.
- Prepare retention-related data.
- Prepare dashboard datasets.

Day 4–7

- Develop FastAPI application.
- Implement /predict.
- Implement /batch_predict.
- Implement /health.
- Test API using Swagger UI.

WEEK 4 – Visualization & Deployment

Day 1–3

- Prepare Metabase dashboard datasets.
- Develop Churn & LTV Overview.
- Develop Churn Risk Dashboard.
- Develop LTV Segmentation Dashboard.

Day 4–5

- Create KPI cards.
- Create charts and visualizations.
- Validate dashboard results.
- Analyze churn and LTV insights.

Day 6

- Containerize FastAPI using Docker.
- Start the Docker container.
- Verify the API on port 8000.
- Test Swagger endpoints.

Day 7

- Perform final API testing.
- Verify /predict.
- Verify /batch_predict.
- Verify /health.
- Verify Metabase dashboards.

- Prepare final project documentation.

26. API TESTING SUMMARY

component	Result
FastAPI	successful
Swagger UI	successful
/predict	200 ok
/batch_predict	200 ok
/health	200 Ok
Docker Containers	Running
Postgresql	Used
SqlAlchemy	Used
metabase	Completed
Churn &Ltv Overview	Completed
Churn Risk	Completed
LTV segmentation	Completed

27. KEY PROJECT OUTCOMES

The project successfully demonstrates:

- Customer churn analysis.
- Customer churn prediction.
- Churn probability estimation.
- Single-customer prediction.
- Batch customer prediction.
- Customer Lifetime Value analysis.
- LTV segmentation.
- Churn-risk analysis.
- Retention-related analysis.
- PostgreSQL database usage.
- SQLAlchemy database connectivity.
- FastAPI REST API development.
- Swagger API testing.
- Docker deployment.
- Metabase business intelligence.

KPI cards.

- Customer churn charts.
- LTV segmentation charts.
- Customer risk visualizations.

28. EXPECTED BUSINESS IMPACT

The solution can help telecommunications and subscription-based businesses to:

- Identify customers with higher churn risk.
- Support proactive customer retention.
- Understand customer behavior.
- Identify high-value customer groups.
- Prioritize retention activities.
- Support LTV-based customer segmentation.
- Improve data-driven decision-making.
- Optimize customer retention strategies.

29. CONCLUSION

The Customer Churn Prediction & Lifetime Value (LTV) Engine is an end-to-end data analytics and predictive analytics project designed to transform customer data into useful business insights.

The project starts with the Telco Customer Churn Dataset and performs data preparation, exploratory analysis, churn analysis, LTV analysis, and customer segmentation.

PostgreSQL and SQLAlchemy provide the database layer, while FastAPI provides REST API functionality for single and batch churn prediction.

Docker is used to containerize and run the API application.

Metabase provides three business intelligence dashboards:

1. Churn & LTV Overview
2. Churn Risk
3. LTV Segmentation

The project demonstrates the complete workflow from raw customer data to analytics, prediction, API deployment, and business intelligence visualization.

The final solution can support organizations in identifying churn-risk customers, understanding customer value, and making data-driven customer retention decisions.

FINAL PROJECT FLOW

Dataset



Data Cleaning



Exploratory Data Analysis

↓

Feature Engineering

↓

Churn Prediction

↓

LTV Analysis & Segmentation

↓

PostgreSQL + SQLAlchemy

↓

FastAPI

↓

Swagger API Testing

↓

Docker Deployment

↓

Metabase

↓

Churn & LTV Overview

↓

Churn Risk

↓

LTV Segmentation

↓

Business Insights

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